

CCAT AI Coach - Completed Pre-Search

1. Domain Selection

Domain: EdTech / Employment Assessment. Use cases: personalized CCAT tutoring, adaptive practice tests, diagnostics, pacing coaching, analytics. Verification: deterministic math/scoring, reviewed reasoning. Data: proprietary CCAT-style questions, user telemetry, cognitive science literature.

2. Scale & Performance

MVP: 500–2,000 users; Year 1: 20k–50k. Tutor latency <2s, analytics <1s, generation <5s. 500 concurrent users. Target inference cost <\$0.10/session.

3. Reliability

Critical: correct scoring and math. High: accurate explanations. Verification pipeline for solutions. Human review only for content moderation.

4. Team & Skills

Small AI-first team using AI builders, moderate software engineering, LangGraph familiarity to be developed.

5. Agent Framework

LangGraph multi-agent: Tutor, Diagnostic, Strategy Coach, Question Generator, Analytics.

6. LLM Selection

Primary: GPT-5. Future: smaller model for inexpensive question generation.

7. Tool Design

Question bank, adaptive generator, timer, score calculator, verifier, analytics, user history, difficulty estimator.

8. Observability

LangSmith + PostHog. Metrics: score, pacing, solve time, skips, confidence, mastery, cost.

9. Evaluation

Ground truth from verified solutions and pilot users. Measure score improvement, prediction accuracy, explanation ratings.

10. Verification

Math checked deterministically; scoring deterministic; explanations cross-checked before display.

11. Failure Modes

Fallback to cached questions, offline practice, graceful degradation, ambiguity explanations.

12. Security

Minimal PII, encrypted storage, secure API keys, audit logs.

13. Testing

Unit, integration, adversarial, regression, timing accuracy.

14. Open Source

Release evaluation framework and prompts; keep personalization engine proprietary.

15. Deployment

Next.js frontend, Supabase backend, OpenAI APIs, LangSmith monitoring, CI/CD.

16. Iteration

Collect telemetry and feedback, biweekly eval-driven improvements, prioritize highest-impact features.